Orientation: the minimal branch

One orphan branch holds the source, tests and recipes for the release and fast configurations; everything else is in the archive tag.

WHERE THINGS ARE

+ Repo nasa-nccs-hpda/or0cky, default branch minimal-port-20261016. Archive: tag archive/full-fidelity-port-20261016 (scoping notes, drafts, run records).
+ Read in order: README.md, rocke3d_jax/README.md, STATUS_20261016.md, DATA_AND_PATHS.md.
+ Data (ff_data, the ModelE tree, binaries) lives outside the repo.

SIZE

+ fullfidelity/: 317 Python modules (about 76,600 lines) plus 135 test files; socrates_spike/: 16 modules.
+ gpu/: run_variants.sh, summarize_variants.py, benchmark and gate scripts, environment file.
+ Reports/: acceptance criteria, release and speed recipes, repeatability report.

Entry point	What it does
gpu_day_run.py -> d212_day.py	GPU day run (54 or 108 steps); --rad native or server
gpu_step_run.py / d193_day.py	6-step GPU gate / the day loop under it
cpu_day_run_imf.py	CPU day with Intel libimf (dry-run only so far)
d207_score.py, multiday_score.py, atm_day_report.py	Scoring against the five one-ulp members; defines within / near / beyond
run_all_tests_sharded.sh, run_all_tests_flags_on.sh	Regression subset, flags off / nine flags on

Sources: README.md; rocke3d_jax/README.md; STATUS_20261016 s1; NEW_BRANCH_BRIEF_20261016 s3 (archive)

Anatomy of a coupled step

A mixed host/device program: the release path is all flags off; the fast path swaps stages through opt-in flags.

STATE + SETUP
restart, five step-0 dump files, providers

>

ATMOSPHERE
dyn_step_jax2.py
jax_p1_condse.py (CONDSE)

>

SURFACE + LAND
surface tiles, PBL, GHY Ent, daily update

>

OCEAN + ICE
ocean_chain_io.py
sea ice, lakes, pole

>

RADIATION
socrates_spike/
radia_native.py

RELEASE CONFIGURATION

+ jax_coupled.py Coupled.step drives the stages on device; atm_step.py is the NumPy chain used for C1 consistency.
+ Native radiation, standalone flags (ROCKE_TPL_OWN etc.) set by the runner; no speed flags.
+ Needs XLA flags --xla_disable_hlo_passes=algsimp,reshape-mover on CPU (cold compile 30x faster, bitwise identical).

FAST CONFIGURATION (nine flags)

+ MSTCNV on the host in a persistent Numba worker (ROCKE_MSTCNV_PY), libimf exp/pow, batched host I/O.
+ QDYN vmap, pole columns fast + async, sequential-sum blocks of 32, glue jit, Ent batching.
+ Flags default off; ENT_BATCH is not bitwise equal to the level-by-level version.

Input boundary: nov26 restart plus five step-0 Fortran dumps; no per-step records are read in the standalone-capable path.

Sources: JAX_COVERAGE_MATRIX (archive); SPEED_RUN_RECIPE_20261013; RELEASE_NATIVE_RADIATION_20261012; ENVIRONMENT.md; LESSONS: XLA flag fix D182

Running it: one command, seven SLURM jobs

gpu/run_variants.sh submits everything and a dependent summary job; it prints one table, gpu_runs/<name>/SUMMARY.md.

Job	Where	What
pair_release, pair_fast	exclusive V100, 8 CPUs, 64 GB	Fortran x3 on one core, then JAX, 108 steps, same node (timing pair)
fid_release, fid_fast	V100, shared node ok	54-step day, scored against the five members by d207_score.py
tests_off, tests_on	CPU	Sharded regression with speed flags off / nine flags on; 2 isolated files
summary	CPU, runs after the others end	summarize_variants.py writes SUMMARY.md and score_<kind>.md

SETUP (DATA_AND_PATHS.md s4)

+ Export FF_DATA, MODELE_PROD_INPUT (the complete ModelE_Support/prod_input_files), ROCKE_SOCRATES_DIR, FORTRAN_HUGE_SPACE, FORTRAN_RUNS, ROCKE_MSTCNV_PY, CONDA_BIN.
+ Optional: ROCKE_ENV (name or absolute path), SBATCH_HOST=gpulogin1, ONLY='fid_fast tests_off', DRY_RUN=1.
+ Pre-flight checks exit with code 2 and name what is missing.

EXPECTED RESULT (RUN E, 18 OCT)

+ All seven jobs exit 0 in 38 minutes from submission (queue time extra).
+ Fidelity worst ratio (steps >= 3): release QCI 2.37, fast QCI 2.47.
+ Tests: 3,288 passed, 0 failed, 40 skipped in both configurations.
+ Fortran 251-254 s per run; JAX step 0 about 940-1,090 s, step 1 96-104 s.

forest204 has no SLURM client: set SBATCH_HOST=gpulogin1. Set PYTHONNOUSERSITE=1 so a personal ~/.local cannot change results.

Sources: gpu/run_variants.sh; gpu/summarize_variants.py; DATA_AND_PATHS s4; STATUS_20261016 s5 (run e)

Environment, data and cluster facts

Everything outside the repository, and the traps that cost the most time.

ENVIRONMENTS (READ-ONLY, GROUP ilab)

+ JAX: /explore/nobackup/projects/ilab/conda/envs/rocke3d-jax (Python 3.10, jax 0.5.3, NumPy 2.2.4, mpmath 1.3.0), made by conda-pack; passed the 6-step V100 gate bitwise.
+ Numba worker (ROCKE_MSTCNV_PY): the ilab pytorch-kernel-copy_20260408 Python (3.12, numba 0.60, NumPy 2.0.2).
+ Rebuild from gpu/environment-graphcast-env.yml is untested (package server rate limit).

TEST HOSTS (XEON GOLD 6248, AVX-512)

+ JAX_PLATFORMS=cpu, or a job landing on a GPU node fails to start CUDA.
+ XLA_FLAGS='--xla_cpu_max_isa=AVX --xla_disable_hlo_passes=algsimp,reshape-mover'.
+ NPY_DISABLE_CPU_FEATURES for the AVX-512 set: NumPy pow differs in the last bit otherwise (probe: 16 of 83 failed).
+ MODELE_SRC must be set (3 tests skip without it).

TRAPS

+ MODELE_PROD_INPUT must be the complete prod_input_files; the in-repo copy is partial and a day run then fails early behind a misleading 'NoneType ... calls' error.
+ Two tests (test_daily_lake_hook, test_d187_imf_p1) must run in their own process; test_libimf_fused is order-dependent. test_jax_state collects no tests (rc=5 is expected).
+ The data is group-readable under the author's directories (never o+r); do not copy it into home directories without asking.

Sources: DATA_AND_PATHS; fullfidelity/qa_20261016/ENVIRONMENT.md; STATUS_20261016 s5; run_variants.sh

Benchmark arithmetic and what the speed claim covers

Separate steady-state throughput from start-up; compare only equal windows.

THE MATCHED PAIR

- One exclusive V100 node: Fortran on one core (x3), then JAX, same nov26 restart, 108 steps (2 model days).
- Archived fast pair (job 37941875): Fortran main loop 246.7 / 246.6 s (2.285 s/step); JAX steps 2-107: 55.6 s (0.524 s/step).
- Branch runs: Fortran whole process 251-254 s (about 2.35 s/step, includes start-up).

WHAT IT DOES NOT INCLUDE

- Step 0: 940-1,090 s; step 1: 96-104 s (compile); no persistent XLA cache.
- Crossover with Fortran: (940 + 100) / (2.285 - step time) = about 590 steps at 0.524 s, about 700 at 0.80 s, i.e. 12-15 days at 48 steps/day. Arithmetic, not measured.
- Release: 1.7-2.2 s/step, parity at best.

Say '2.7-4.4x steady state on one fast configuration (V100, nov26)'. Do not say 'the model runs 4.4x faster': start-up dominates a 108-step run and the long-run workload is untested.

Run	Mean s/step	Median	Note
Fast, archived matched (37941875)	0.524	0.456	steps 2-107
Fast, best / worst of 9 runs	0.508 / 0.841	0.453 / 0.789	tag and branch code, exclusive nodes
Release, branch (run e)	1.729	1.689	gpu011

Sources: SPEED_STEP_TIME_LOG_20261012 s13 (archive); STATUS_20261016 s5; RELEASE_NATIVE_RADIATION_20261012; gpu/summarize_variants.py

Why the fast configuration is faster

A bundle of changes aimed at measured bottlenecks, not one kernel optimization.

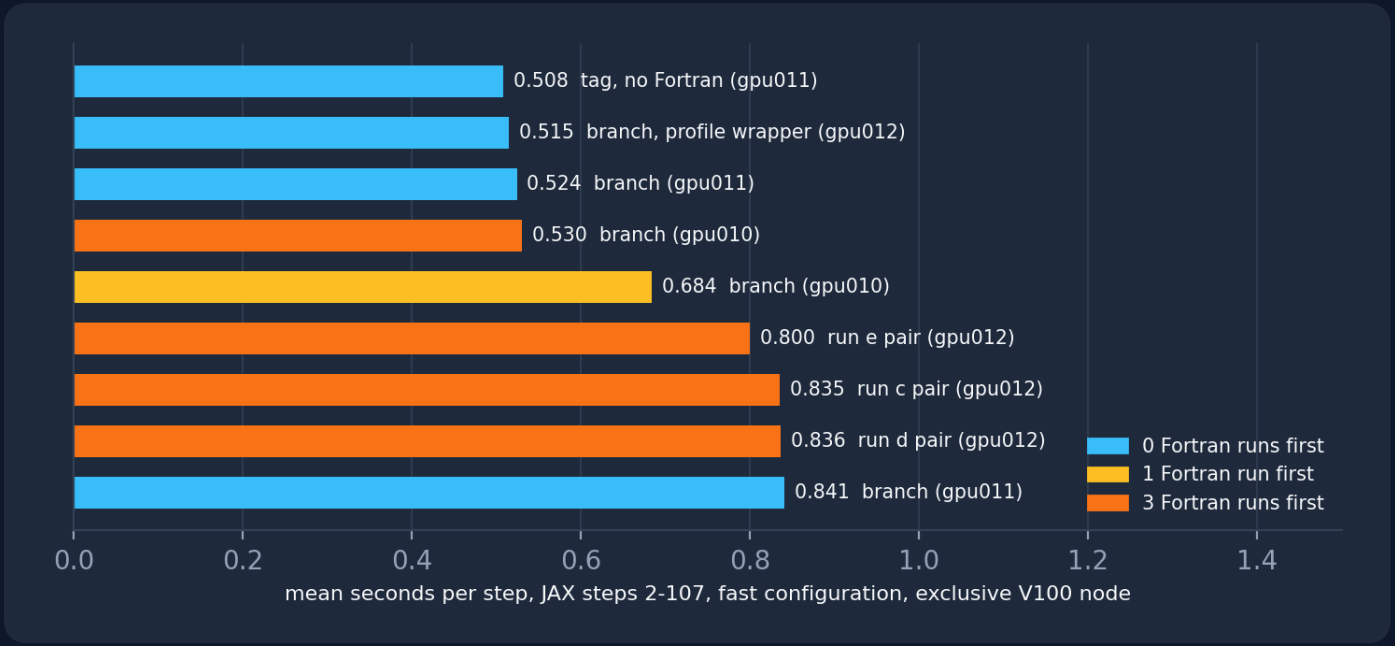
Stage / change	What changed	Evidence / caveat
Convection (MSTCNV)	Host Numba worker, Intel libimf exp/pow, batched host I/O	57 of 57 fields equal to the NumPy-libimf port on captured columns; needs a second Python environment
Dynamics, ocean, pole	QDYN vmap, pole columns fast + async, sequential-sum blocks of 32	Targeted checks; scan unrolling tried and rejected (slower)
Land / Ent	Batch leaf evaluations across qsimp levels and sun/shade	Ent call 5.39 -> 2.37 ms; not bitwise equal (1e-15 relative)
Host glue	Template-state glue as one jit; one batched device_get/put	MSTCNV stage 0.087 -> 0.071 s
Radiation	SOCRATES JAX inside the coupled step	About 0.17 s per call on a V100; median radiation step 10.0 -> 1.9 s

Held step 1.71 s (baseline) -> 0.385-0.402 s in the archived profile runs; remaining held-step stages: surface 0.086 s, MSTCNV 0.071 s, dynamics 0.065 s, ocean_b 0.048 s, plus about 0.16 s on radiation steps. XLA command buffers and larger scan unrolls were slower and were not adopted.

Sources: SPEED_STEP_TIME_LOG_20261012 s11-13 (archive); SPEED_RUN_RECIPE_20261013; SOCRATES_R2_DUMPS s12 (archive)

The open puzzle: the same fast run takes 0.52 s or 0.84 s

Nine exclusive-node runs of one configuration; colour = Fortran runs before JAX on that node.



WHAT WAS RULED OUT

- The code: tag and branch both give 0.51-0.53 s, and the held step is 0.365 s on both under the profile wrapper.
- The node: gpu011 gave fast, fast and slow; gpu012 slow x3, fast x1.
- Fortran before JAX: an early 0/1/3 pattern was reversed by two simultaneous jobs.
- Memory, NUMA, GPU clocks, throttling, other processes: no difference sampled.

Cause not identified. In slow runs the whole step-time distribution shifts up (held-step median 0.78 s against 0.46 s); the fastest step is 0.43 s in every run. Quote the range.

Suggested next step: instrument a slow and a fast job with per-stage timers (ROCKE_DYN_TIMERS, ROCKE_SURF_TIMERS, ROCKE_MSTCNV_HOST_TIMING) on exclusive nodes started together, and compare which stage moves. The profile wrapper itself adds 0.09-0.34 s of file I/O per step; do not compare its step means.

Sources: STATUS_20261016 s5 (jobs 37943471-37943488); SPEED_RUN_RECIPE_20261013 (diagnostic flags)

Tests, gates and open work

What was checked on this branch, what was changed on purpose, and what is left.

GATES (16-18 OCT)

+ Kept files identical to the archive except about 110 path-scrubbed files and a few changed on purpose (listed in STATUS s2).
+ 6-step V100 run bitwise identical to jobs 37942390, 37942591, 37943287; also from the shared environment (job 37943470).
+ Scoring reproduces the review-package table (QCI 2.37 at step 3).
+ Regression: 3,288 passed, 0 failed, 40 skipped (flags off and on); 23 of 40 skips traced to declared causes, 17 untraced.

OPEN WORK (PRIORITY ORDER)

+ Second-person reproduction from a fresh clone (nobody but the author has run it).
+ Explain the fast step-time variation (previous slide).
+ CPU libimf day; localize the native-radiation repeat differences; trace the 17 skips.
+ Rebuild the environment from its file; package the Numba worker.
+ Fix the 30-day failure (qdynam_do_z_extra) and run beyond two days; other start dates.

Report both strict and relaxed fidelity results every time; label the radiation actually used (--rad native or server); never state a result the SUMMARY.md of a run does not show.

Records: STATUS_20261016.md is the record of what was verified; Reports/ACCEPTANCE_CRITERIA.md is the owner-approved rule set (amendments signed off 2026-10-17); the archive tag holds the full ledger (FULL_FIDELITY_DELTAS.md, D1-D198).

Sources: STATUS_20261016 s2, s4-6; ACCEPTANCE_CRITERIA; run_all_tests_sharded.sh; QA_HANDOFF_20261016 (archive)